Data Integrity Report

Experiment:
Therapeutic Efficacy of TOUCAN-MRSA 20mg-kg-day_PoC Study

Date:28.05.2020, 20.06.2020, 09.07.2020

By: Baris Ata Borsa & Frank Hernandez

APPROVED by Frank J. Hernandez, 2026-03-06 15:03

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Experimental Conditions

Experiment Overview: General Information

- **Study Title:** Therapeutic Efficacy of TOUCAN-MRSA 20mg-kg-day_PoC Study

Experiment Date and time: 28.05.2020, 20.06.2020, 09.07.2020

- **Laboratory:** Lab U at AB3 facility of CBR (Campus US, LiU) and BSL-2 microbiology lab at 2E:686 (Campus Valla, LiU).

- **Protocols:** *SOP-TL-THIGINF* for animal work, *SOP-TL-CULT* for bacteria cultures, *SOP-TL-CFU* for quantitative tissue cultures.

- **Ethical permission:** Dnr 14980-2023 ID5830

- **Researcher:** Baris Ata Borsa & Frank Hernandez

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Experiment Overview: Study Design

- **Bacteria strain:** *Staphylococcus aureus* ATCC BAA-1766
- **Mouse:** C57BL/6 (4-6 weeks, female)
- **Drug substances:**
  - TOUCAN-MRSA (iv),
  - PBS (iv, as placebo)

- **Experimental groups (n=4):**
  - Non-infected / PBS-treated (q6h, 100 ul)
  - Infected / PBS-treated (q6h, 100 ul)
  - Infected / TOUCAN-MRSA treated (equal to 20 mg/kg/day floxuridine, q6h in 100 ul)
  - Infected / floxuridine treated (20 mg/kg/day, q6h in 100 ul)

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Experimental Schedule

XXX	XXX	XXX	XXX	XXX
Bacteria culture on agar plate (BSL-2)	Preparation of the drug substances (BSL-2)	Bacteria Inoculation at 09:00 h (Lab U)	Drug administration: 4th dose at 05:00 h (Lab U)	Analysis of the culture results
	Verification of the experiment conditions at Lab U (anesthesia equipment, surgical equipment, lighting, trimmer, camera etc.)	Drug administration: 1st dose at 11:00 h (Lab U)	Euthanasia and dissections/sampling at 09:00 h (Lab U)	
	Broth culture of bacteria at 20:00 h (BSL-2)	Drug administration: 2nd dose at 17:00 h (Lab U)	Processing (homogenization/dilution) and bacteriological analysis (agar cultures for CFU analysis) of the samples at 13:00 h (BSL-2)	
		Drug administration: 3rd dose at 23:00 h (Lab U)		

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Data& Analysis – Bacteriological analysis

Group	Code	CFU per gram	Average CFU per gram	% CFU decrease vs placebo	Log decrease vs placebo
Non-infected / PBS-treated	**NR**	0	0	**NA**	**NA**
	NL	0			
	NN	0			
	NB	0			
Infected / PBS-treated	**PR**	66666667	136174824	**NA**	**NA**
	PL	89552239			
	PN	145833333			
	PB	242647059			
Infected / TOUCAN-MRSA treated	**TR**	72034	212182	**99,84**	**2,8**
	TL	158103			
	TN	259740			
	TB	358852			
Infected / floxuridine treated	**FR**	647058824	625624616	**0**	**0**
	FL	757575758			
	FN	491803279			
	FB	606060606			

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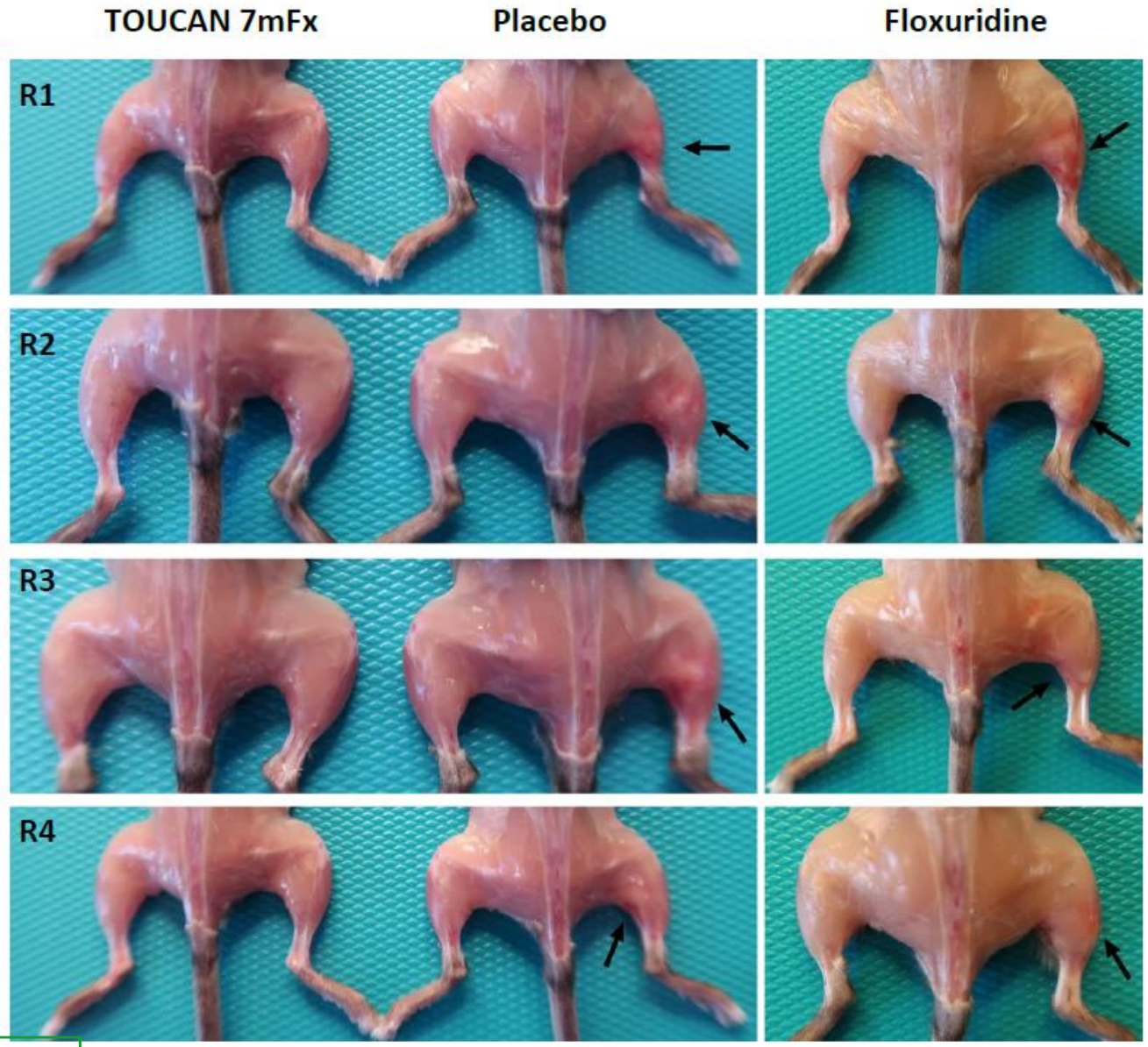
Data & Analysis – Animal pictures



Gross replicate photographs of TOUCAN 7mFx, Placebo and Floxuridine groups.

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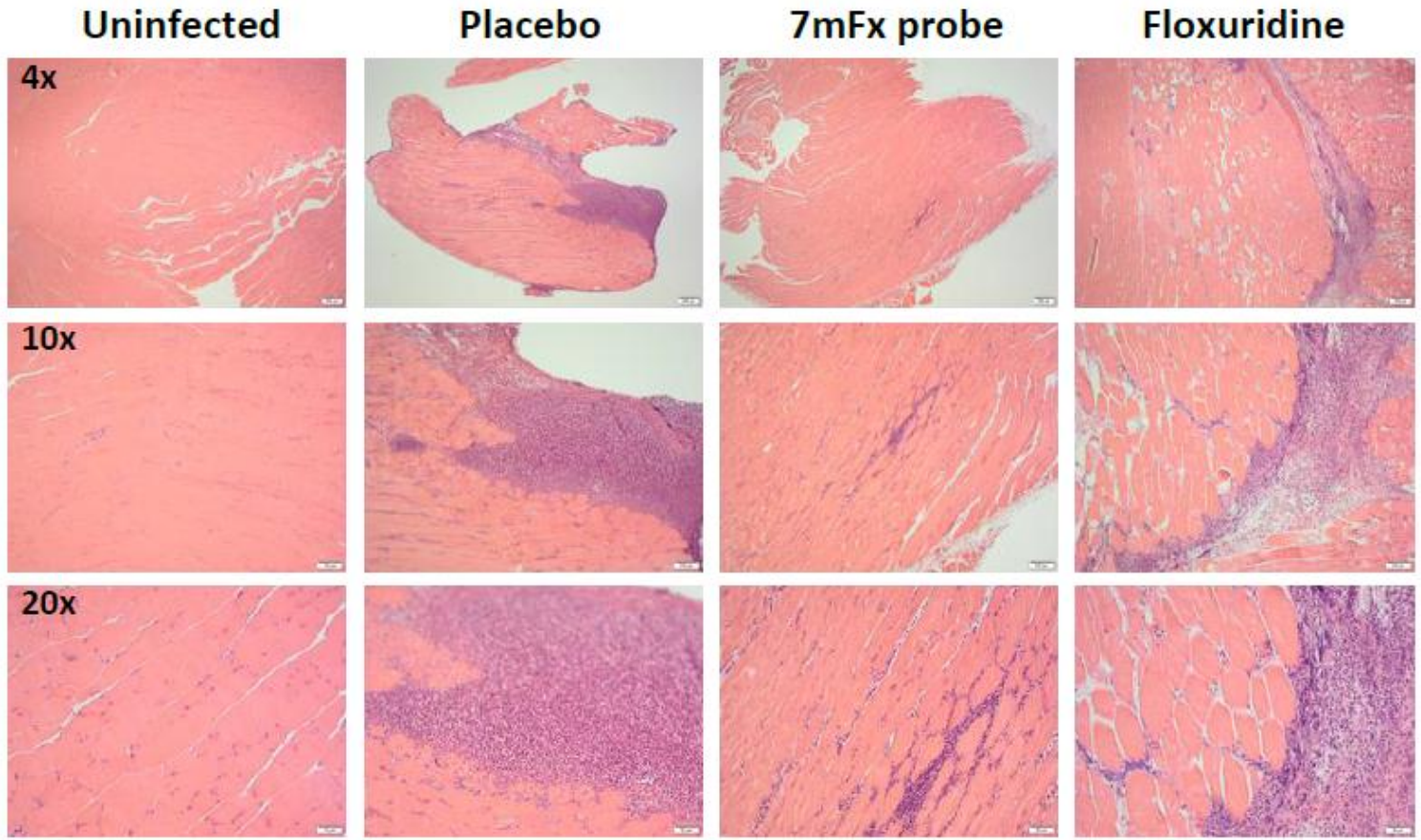
Data & Analysis – Histopathology



Representative histopathological images of uninfected (control left leg) and infected (right leg) mouse tissues using different optical objectives (4x, 10x and 20x).

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Data & Analysis – Blood counts

Blood parameters	Units	Uninfected control		Placebo		TOUCAN 7mFx		Floxuridine	
		AVG	STD	AVG	STD	AVG	STD	AVG	STD
RBC	x10^12/L	9,3	0,2	9,5	0,3	9,3	0,2	8,6	0,4
HCT	%	44,0	0,1	46,3	1,5	44,5	0,8	40,8	2,1
HGB	g/dl	13,8	0,3	14,2	0,5	14,0	0,3	12,8	0,6
MCV	fL	47,5	0,4	48,5	0,5	47,9	0,7	47,4	0,6
MCH	pg	14,8	0,1	14,9	0,2	15,0	0,2	14,8	0,4
MCHC	g/dl	31,3	0,4	30,7	0,5	31,4	0,5	31,5	0,3
RET%	%	3,0	0,3	2,2	2,0	2,8	0,4	4,5	0,4
RET	K/µL	279,7	22,3	298,4	45,3	257,7	37,4	384,9	19,6
WBC	10^9/L	4,3	1,2	2,6	1,1	2,4	0,3	2,2	0,3
NE%	%	9,8	0,9	29,0	11,9	14,1	1,9	36,1	9,5
LYM%	%	83,6	0,8	53,6	6,3	68,6	6,9	48,9	10,7
MON%	%	4,0	0,6	16,0	5,9	15,7	9,2	13,5	12,7
EO%	%	2,3	1,2	1,3	0,7	1,3	0,4	1,0	0,6
BAS%	%	0,3	0,3	0,1	0,2	0,3	0,2	0,5	0,6
NE	10^9/L	0,4	0,1	0,7	0,2	0,3	0,1	0,7	0,3
LYM	10^9/L	3,6	1,0	1,4	0,7	1,6	0,4	1,1	0,4
MON	10^9/L	0,2	0,1	0,4	0,3	0,4	0,2	0,3	0,2
EO	10^9/L	0,1	0,1	0,0	0,0	0,0	0,0	0,0	0,0
BAS	10^9/L	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0
PLT	K/µL	743,0	20,9	820,3	24,2	630,0	165,6	599,3	210,3

Blood counts from uninfected/untreated and infected/treated mouse groups (mean ± SD, n = 3).

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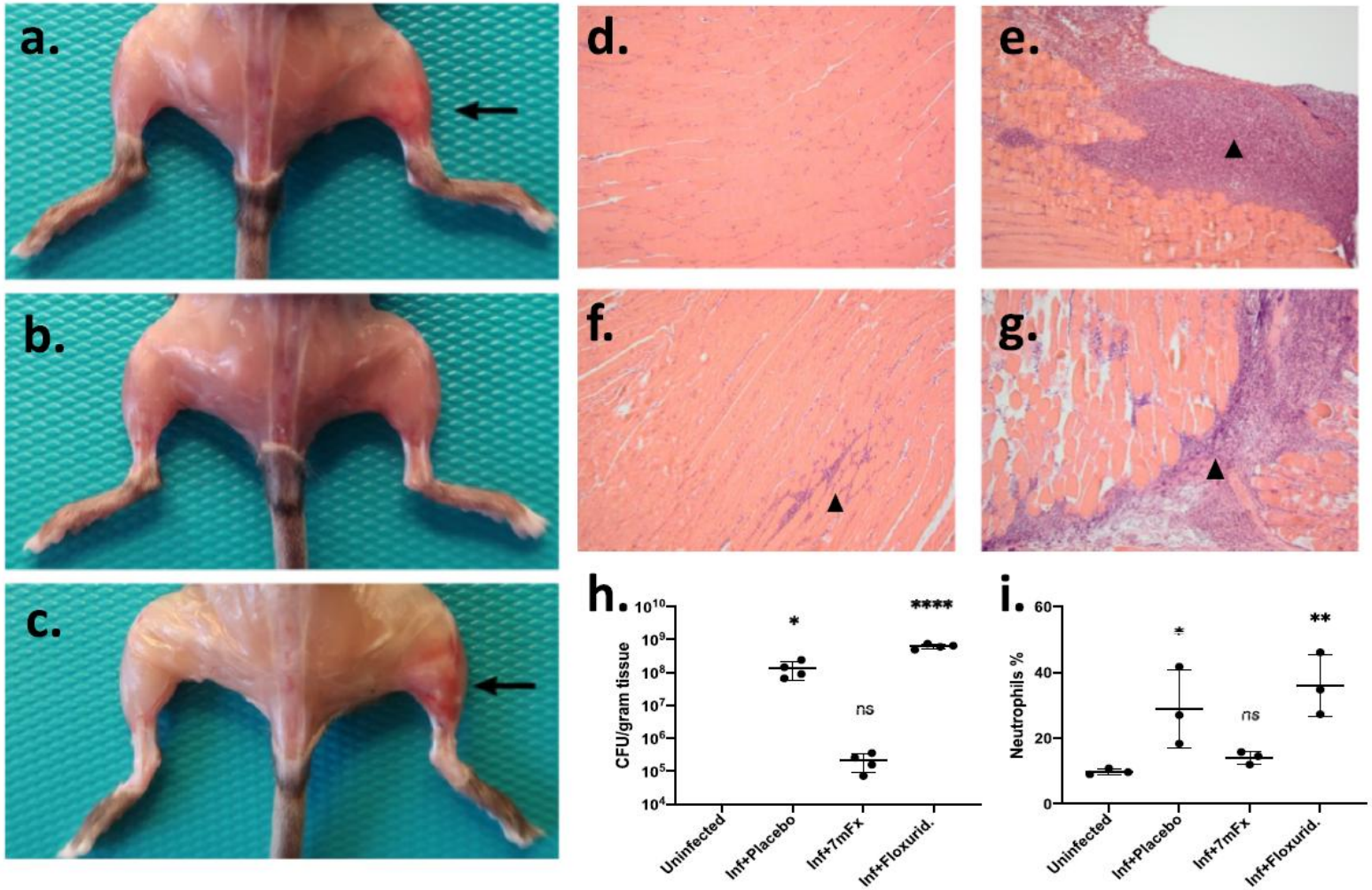
Experimental results



Therapeutic efficacy of TOUCANs in mice. Representative gross photographs of mice groups treated with a) PBS, b) TOUCAN 7mFx and c) floxuridine. Histopathological evaluation of mouse leg tissues of d) uninfected/untreated, e) infected treated with placebo (PBS), f) infected treated with TOUCAN 7mFx and g) infected treated with floxuridine. All photographs are representative of at least 4 mice per group. h) Bacteria counts per gram tissue of uninfected and treated infected legs reported as CFU/g. i) Percentage blood neutrophils of uninfected and infected mice. Statistical analysis for the bacterial counts and neutrophils was performed with a one-way ANOVA using Dunnett's multiple comparison test. Data is presented as mean ± SD, $n = 4$ for CFUs and n = 3 for neutrophils. Statistical significance *vs.* uninfected leg control is denoted by **** $p < 0.0001$, * $p < 0.0332$ and ns =0.1234 stands for no significant difference when compared to uninfected leg.

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Date: 2024-12-22
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